

Microelectronics

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1 Microelectronics

I Signals

i Signal

A signal is a function that conveys information about a phenomenon. Signal processing is usually most conveniently performed by electronic systems, and voltage and current signals are studied here.

ii Transducer

A transducer is a device that convert a signal to a voltage signal.

iii Signal source

A signal source is typically represented in the Thévenin form, a voltage source in a series connection with a resistance, or the Norton form, a current source in a parallel connection with a resistance.

iv Symbol convention

- Lowercase character with uppercase subscript means total signal, e.g. $i_C(t)$.
- Lowercase character with lowercase subscript means AC signal, e.g. $i_c(t)$.
- Uppercase character with uppercase subscript means DC signal, e.g. I_C .
- Uppercase character with lowercase subscript means peak signal, e.g. I_c .

Their relation is

$$i_C(t) = i_c(t) + I_C$$
$$i_c(t) = I_c \sin(\omega t + \varphi)$$

II Amplifier

i Amplifier

An electronic device which amplifies an electrical signal by a scale called amplifier gain A and optionally convert voltage signal to current signal or vice versa.

ii Circuit model of voltage amplifier

A voltage amplifier draws current from a source v_s with internal resistance R_s with input resistance R_i and voltage v_{in} across it and supplies $v_c = \mu v_{in}$ to a load network with output resistance R_o and load R_L .

$$\frac{v_{in}}{v_s} = \frac{R_i}{R_s + R_i}$$

$$\frac{v_c}{v_{in}} = \mu$$

$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Voltage gain:

$$A_v = \frac{v_{out}}{v_{in}} = \mu \frac{R_i R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\mu|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \ll R_L, \quad A_v \approx \mu.$$

iii Circuit model of current amplifier

A current amplifier draws current from a current source i_s with internal resistance R_s in parallel with input resistance R_i and current i_{in} through it and supplies $i_c = \beta i_{in}$ by CCCS to a load network with output resistance R_o in parallel and load R_L .

$$\frac{i_{in}}{i_s} = \frac{R_s}{R_s + R_i}$$

$$\frac{i_c}{i_{in}} = \beta$$

$$\frac{i_{out}}{i_c} = \frac{R_o}{R_o + R_L}$$

Current gain:

$$A_c = \frac{i_{out}}{i_{in}} = \beta \frac{R_s R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |\beta|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \gg R_L, \quad A_c \approx \beta.$$

iv Circuit model of transconductance amplifier

A transconductance amplifier draws current from a voltage source v_s with internal resistance R_s in series with input resistance R_i and current i_{in} through it and supplies $i_c = g v_{in}$ by VCCS to a load network with output resistance R_o in parallel and load R_L .

$$\frac{v_{in}}{v_s} = \frac{R_i}{R_s + R_i}$$

$$\frac{i_c}{v_{in}} = g$$

$$\frac{i_{out}}{i_c} = \frac{R_o}{R_o + R_L}$$

Gain:

$$A_g = \frac{i_{out}}{v_{in}} = g \frac{R_i R_o}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |g|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \gg R_s, \quad R_o \gg R_L, \quad A_g \approx g.$$

v Circuit model of transresistance amplifier

A transresistance amplifier draws current from a current source i_s with internal resistance R_s in parallel with input resistance R_i and current i_{in} through it and supplies $v_c = r i_{in}$ by CCVS to a load network with output resistance R_o in series and load R_L .

$$\frac{i_{in}}{i_s} = \frac{R_s}{R_s + R_i}$$

$$\frac{v_c}{i_{in}} = r$$

$$\frac{v_{out}}{v_c} = \frac{R_L}{R_o + R_L}$$

Gain:

$$A_r = \frac{v_{out}}{i_{in}} = \beta \frac{R_s R_L}{(R_s + R_i)(R_o + R_L)}$$

$|A| < |r|$ due to loading effects.

A good amplifier requires small loading effect:

$$R_i \ll R_s, \quad R_o \ll R_L, \quad A_c \approx r.$$

vi Amplifier gain in decibels (dB)

A power gain A_p in dB is defined as

$$G_p = 10 \log_{10}(A_p).$$

A voltage gain A_v in dB is defined as

$$G_v = 20 \log_{10}(A_v).$$

A current gain A_c in dB is defined as

$$G_c = 20 \log_{10}(A_c).$$

vii Amplifiers in cascade

Output of an amplifier as input of another amplifier.